Clinical Genomics Evidence & Deep Research Synthesis: Daniel Ehrle

Patient / Cohort Identifier: `Daniel_Ehrle` | **Report Date:** October 13, 2026

Target Audience: Medical Geneticist · Subspecialty Investigator · Primary Care Physician · Clinical Bioinformatician

Sequencing Depth & Reference: BAM Aligned Mean Depth **48.8x** on GRCh38.p14

Orientation: What We Are Covering

This clinical genomics research dossier evaluates **1407 prioritized variants across 1002 clinical genes** and **41 pharmacogenes (377 screened medications)** for **Daniel Ehrle** (`Daniel_Ehrle`), backed by **48.8x** BAM mean sequencing depth. This synthesis separates **Mechanistic Causality** (coding loss-of-function, canonical splice disruption, 3D active-site destabilization) from **Statistical Correlation** (GWAS linkage disequilibrium proxies and common polygenic modifiers), audits **Intra-Gene Cis/Trans Phasing** across multi-variant genes, and provides a **Biomarker Ground-Truthing & Longitudinal Futures Roadmap**.

Body

1. Primary Mechanistic & High-Priority Clinical Findings (With Locus Depth, VAF & Confidence)

Gene	Variant & Coordinate (GRCh38)	Zygosity & Phase	Locus DP & VAF	ClinVar Status	CADD (Phred) / REVEL	AlphaGenome AVI (Phred) & Top Driver	Confidence
CBLIF	`c.79+1G>A` (`chr11:59845374`)	Heterozygous (Maternal)	`49x` / `0.47`	**Pathogenic**	32.0 / —	33.9 (Splicing)	**0.95**
GJB2	`p.Met34Thr` (`chr13:20189481`)	Heterozygous (Unphased)	`57x` / `0.63`	**Pathogenic**	20.9 / 0.70	23.6 (Cactus)	**0.99**
VDR	`c.-1172A>G` (`chr12:47906043`)	Homozygous (N/A)	`49x` / `1.00`	**Likely pathogenic**	5.4 / —	2.1 (DNASE-seq)	**0.85**
F5	`p.Arg534Gln` (`chr1:169549811`)	Heterozygous (Unphased)	`58x` / `0.62`	**drug response**	27.9 / AM:0.20	23.6 (Cactus)	**0.99**
PEX6	`p.Val788Met` (`chr6:42966044`)	Heterozygous (Unphased)	`48x` / `0.58`	**Conflicting classifications of pathogenicity**	35.0 / 0.92	32.3 (Cactus)	**0.88**
GJB3	`p.Arg75Cys` (`chr1:34784985`)	Heterozygous (Maternal)	`38x` / `0.37`	**Conflicting classifications of pathogenicity**	26.2 / 0.94	32.4 (AlphaMissense)	**0.88**
SCN4A	`p.His599Arg` (`chr17:63961242`)	Heterozygous (Unphased)	`56x` / `0.59`	**Conflicting classifications of pathogenicity**	24.4 / 0.93	27.8 (AlphaMissense)	**0.88**
C19orf12	`p.Gly42Arg` (`chr19:29708290`)	Heterozygous (Maternal)	`52x` / `0.40`	**Conflicting classifications of pathogenicity**	24.3 / 0.78	29.4 (AlphaMissense)	**0.88**
LRP5	`p.Val233Glu` (`chr11:68363758`)	Heterozygous (Unphased)	`69x` / `0.55`	**Uncertain significance**	24.9 / 0.90	25.4 (AlphaMissense)	**0.88**
BLM	`p.Val724Ala` (`chr15:90765392`)	Heterozygous (Unphased)	`39x` / `0.44`	**Uncertain significance**	27.5 / 0.47	26.1 (AlphaMissense)	**0.95**

2. Mechanistic Causality vs. Statistical Correlation Matrix

To prevent clinical over-interpretation of statistical GWAS proxies, cohort findings are stratified across four epistemic tiers:

Epistemic Class	Count in Cohort	Representative Genes	Mechanistic vs. Statistical Distinction & Clinical Rule
Q1 · Confirmed Mechanistic Causality	**3**	CBLIF, GJB2, CYP3A5	**Direct Physical Disruption + Clinical Proof:** Concordant ClinVar P/LP curation backed by high AI functional impact (CADD ≥ 20 / AVI ≥ 20). Eligible for preventive clinical surveillance.
Q2 · Cryptic AI-Predicted Disruptors	**166**	DNAH7, GRIA4, HEXA, PLA2G6, TRAK1, TTN	**Biophysical Disruption Without Clinical Database Entry:** Severe deep-learning predicted splice, chromatin, or 3D folding impact currently VUS/unreviewed in ClinVar. Prioritize for biomarker ground-truthing.
Q3 · Database-Curated / Statistical & PGx Modifiers	**321**	CNTN1, TTN, VDR, ALDH4A1, ANK3, APOB	**Curated Hypomorphic, PGx & Statistical GWAS Loci:** Includes enzymatic drug-response alleles and statistical population modifiers; must **never** trigger invasive monogenic workups.
Q4 · Protective & Resilience Offsets	**9**	ADH1C, CCR5, CASP8, MPO, NOS3, CDKN2B	**Risk-Attenuating & Cardioprotective Alleles:** Curated protective modifiers buffering baseline metabolic or vascular pathway susceptibility.

2.1 High-Priority Cryptic AI Disruptors (AlphaGenome 18-Modality Attribution)

Gene	Variant & Coordinate	Zygosity & Phase	AVI (Phred) & Percentile	Primary Modality Driver	Top SHAP Attribution Weights (`fi_*`)
COL23A1	`c.675+2T>C` (`chr5:178262215`)	Heterozygous (Unphased)	**26.3** (Top 0.23%)	**Splicing**	Splicing: 59.0%, Cactus 241-Way: 20.3%, PhastCons 470-Way: 12.1%
ELOVL7	`p.Ser79Tyr` (`chr5:60771922`)	Heterozygous (Maternal)	**31.4** (Top 0.07%)	**AlphaMissense**	AlphaMissense: 48.4%, Cactus 241-Way: 31.5%, PhastCons 470-Way: 13.9%
IGFBP5	`p.Arg193His` (`chr2:216678221`)	Heterozygous (Unphased)	**26.4** (Top 0.23%)	**AlphaMissense**	AlphaMissense: 39.4%, Cactus 241-Way: 38.9%, PhastCons 470-Way: 14.9%
PTDSS2	`p.Glu232Gly` (`chr11:488272`)	Heterozygous (Unphased)	**25.8** (Top 0.26%)	**AlphaMissense**	AlphaMissense: 48.7%, Cactus 241-Way: 28.4%, PhastCons 470-Way: 16.9%
NSMCE1	`p.Cys231Tyr` (`chr16:27225755`)	Heterozygous (Paternal)	**30.6** (Top 0.09%)	**AlphaMissense**	AlphaMissense: 50.2%, Cactus 241-Way: 32.3%, PhastCons 470-Way: 14.0%
BTN3A2	`c.715+2T>G` (`chr6:26370605`)	Heterozygous (Unphased)	**25.4** (Top 0.29%)	**Splicing**	Splicing: 69.3%, PhastCons 470-Way: 14.6%, CAGE Promoter: 4.1%

3. Intra-Gene Cis / Trans Phase Auditor & Cross-Pathway Epistasis

Parental and read-backed phasing across all multi-variant genes in **Daniel Ehrle** resolved **45 Trans Multi-Variant Genes** (variants on opposite parental homologs; 0 confirmed biallelic pathogenic recessive compound-heterozygotes) and **38 Cis Haplotype Genes** (variants co-inherited on the same parental chromosome, leaving the opposite parental homolog completely wild-type).

Gene	Organ System	Phased Architecture	Variant Count (Mat / Pat)	Max CADD (Phred)	Clinical & Inheritance Interpretation
DHRS3	Heart & Cardiovascular	**TRANS (Biallelic / Compound Het Candidate)**	2 (1M / 1P)	28.1	1 Maternal + 1 Paternal alleles in trans across DHRS3
APOB	Heart & Cardiovascular	**TRANS (Biallelic / Compound Het Candidate)**	10 (2M / 3P)	28.0	2 Maternal + 3 Paternal alleles in trans across APOB
ANK2	Multisystem	**TRANS (Biallelic / Compound Het Candidate)**	3 (1M / 1P)	27.7	1 Maternal + 1 Paternal alleles in trans across ANK2
CACNA1S	Heart & Cardiovascular	**TRANS (Biallelic / Compound Het Candidate)**	12 (4M / 3P)	26.8	4 Maternal + 3 Paternal alleles in trans across CACNA1S
VPS13C	Multisystem	**TRANS (Biallelic / Compound Het Candidate)**	3 (1M / 1P)	25.6	1 Maternal + 1 Paternal alleles in trans across VPS13C
MYOM1	Multisystem	**TRANS (Biallelic / Compound Het Candidate)**	3 (2M / 1P)	25.2	2 Maternal + 1 Paternal alleles in trans across MYOM1
TENM4	Multisystem	**TRANS (Biallelic / Compound Het Candidate)**	4 (1M / 1P)	24.3	1 Maternal + 1 Paternal alleles in trans across TENM4
TTN	Heart & Cardiovascular	**TRANS (Biallelic / Compound Het Candidate)**	4 (2M / 1P)	23.4	2 Maternal + 1 Paternal alleles in trans across TTN
PLEC	Heart & Cardiovascular	**TRANS (Biallelic / Compound Het Candidate)**	3 (1M / 2P)	23.2	1 Maternal + 2 Paternal alleles in trans across PLEC
FLT1	Multisystem	**TRANS (Biallelic / Compound Het Candidate)**	2 (1M / 1P)	22.9	1 Maternal + 1 Paternal alleles in trans across FLT1

4. Actionable Pharmacogenomic (CPIC / DPWG) Medication Matrix

Prescribing Action	Screened Drug Count	Key Affected Medications & Causal Pharmacogenes
🚫 **Avoid / Contraindicated**	**2**	**Combined Oral Contraceptives (Estrogen/Progestin)** (F5), **Hormone Replacement Therapy (Estrogen Hrt)** (F5)
⬇️ **Reduce Dose**	**5**	**Celecoxib (Celebrex)** (CYP2C9), **Phenytoin (Dilantin)** (CYP2C9), **Voriconazole** (CYP2C19/NR1I2/POR/ABCB1), **Warfarin (Coumadin)** (CYP2C9), **Ondansetron** (CYP3A5)
⬆️ **Increase Dose / Rapid Clearance**	**2**	**Omeprazole / Pantoprazole / Lansoprazole** (CYP2C19), **Vardenafil** (CYP3A5)
⚠️ **Use With Caution**	**5**	**4-Hydroxycyclophosphamide** (CYP3A5), **Belinostat** (UGT1A1), **Irinotecan (Camptosar)** (UGT1A1), **Isoniazid** (NAT2/NR1I2), **Atazanavir / Ritonavir** (CYP3A5)
🔍 **Monitor**	**8**	**Atazanavir (Reyataz)** (UGT1A1), **Citalopram (Celexa)** (CYP2C19), **Escitalopram (Lexapro)** (CYP2C19), **Flurbiprofen / Ibuprofen** (CYP2C9), **Sertraline (Zoloft)** (CYP2C19), **Erlotinib** (ABCB1), **Glucose** (CRTC2), **Hdl Cholesterol** (CRTC2)
✅ **Preferred Alternatives**	**81**	**Antihypertensives** (AGT/CLNK/NAMPT/PTPRD), **Atorvastatin (Lipitor)** (SLCO1B1), **Beta Blocking Agents, Selective** (), **Bisphosphonates** (FDPS), **Clopidogrel (Plavix)** (CYP2C19), **Melatonin Receptor Agonists** (ABCB1), **Rosuvastatin (Crestor)** (SLCO1B1), **Selective Serotonin Reuptake Inhibitors** (AFAP1L1/GABRA6/CAP2/FKBP5/ABCB1/MCPH1/TRPM3/ADRA2A/OR52I2)

4.1 Allelic Fraction (VAF) Skewing & Mosaicism / Paralog Uncertainty Audit

Gene	Variant & Coordinate	Locus DP	Observed VAF	Classification	Technical Interpretation
PAK3	`p.Thr247Pro` (`chrX:111163700`)	`15x`	**0.27**	Not reviewed	Skewed VAF (<0.35); rule out clonal mosaicism vs. paralog alignment skew before clinical action.
F5	`p.Thr295Ala` (`chr1:169556715`)	`47x`	**0.34**	Uncertain significance	Skewed VAF (<0.35); rule out clonal mosaicism vs. paralog alignment skew before clinical action.
ALKBH3	`p.Arg70Ter` (`chr11:43884007`)	`50x`	**0.34**	Not reviewed	Skewed VAF (<0.35); rule out clonal mosaicism vs. paralog alignment skew before clinical action.
FBXO41	`p.Glu93Gly` (`chr2:73269353`)	`12x`	**0.17**	Not reviewed	Skewed VAF (<0.35); rule out clonal mosaicism vs. paralog alignment skew before clinical action.
TENM2	`p.Arg1266Gln` (`chr5:168204594`)	`41x`	**0.27**	Not reviewed	Skewed VAF (<0.35); rule out clonal mosaicism vs. paralog alignment skew before clinical action.
TCAIM	`p.Arg217His` (`chr3:44392932`)	`31x`	**0.32**	Uncertain significance	Skewed VAF (<0.35); rule out clonal mosaicism vs. paralog alignment skew before clinical action.

Conclusions: Diagnostic & Epistemic Decision Calculus

Arguments FOR Preventive Surveillance & Genotype-Guided Prescribing

1. **High-Depth Analytical Grounding:** Empirical BAM aligned coverage (**48.8x** mean depth) provides high analytical certainty (`0.98`) across SNVs and small indels.

2. **Actionable Pharmacogenomic Safety:** 22 drug-gene interactions across 41 pharmacogenes provide immediate CPIC/DPWG guidance to avoid toxicity and optimize dosing.

3. **Low-Cost Biomarker Ground-Truthing:** Routine PCP blood panels directly verify physiological homeostasis across the sample's specific prioritized pathways.

Arguments AGAINST Over-Intervention (Why Invasive Testing Is Unwarranted)

1. **Heterozygous Recessive Carrier & Modifier Reserve:** Heterozygous variants affecting or potentially affecting one allele in autosomal recessive or modifier genes and uni-allelic *Cis* haplotypes (38 genes) retain one intact wild-type parental copy; invasive diagnostic procedures or restrictive diets are clinically unwarranted.

2. **Statistical GWAS ≠ Monogenic Causality:** Common polygenic modifiers (AF > 1%) reflect population linkage disequilibrium proxies and must never trigger monogenic disease interventions.

Opportunities: Physiological Ground-Truthing & Longitudinal Futures Roadmap

1. **Primary Care Preventive Blood Panel:** Incorporate the sample-specific venous blood panels and preventive evaluations from Section 2 of the Clinical Brief into the annual PCP physical.

2. **Cryptic Splice & Regulatory Functional Validation (Quadrant 2):** The **166 Cryptic AI Disruptors** exhibiting high AlphaGenome `AVI` scores driven by splicing or RNA-seq modalities can be inspected via the AlphaGenome Ref vs. Alt Track Predictions view.

3. **Automated Annual Re-Annotation & SHA-256 State Fingerprint:** Verified 0 variant reclassifications since prior snapshot (2026-10-13 12:23:02; SHA-256 0f7000325413527a).

Confidence & Uncertainty Assessment

- **Analytical WGS Callset & Phasing Confidence: 0.98** (BAM Mean Depth `48.8x`).
- **CPIC / DPWG Pharmacogenomic Prescribing Confidence: 0.95** (41 pharmacogenes, 377 screened drugs).
- **Monogenic Carrier & Surveillance Confidence: 0.94** (Orthogonal ClinVar, OMIM, CADD, and AlphaGenome agreement).
- **Polygenic / VUS Modifier Penetrance Confidence: 0.70** (Requires laboratory biomarker ground-truthing).

Appendix: Supporting Documentation & Variant Cross-Reference Directory

Gene	Variant	Coordinates (GRCh38)	ClinVar	dbSNP	OMIM	ClinGen	AlphaGenome Atlas
CBLIF	`c.79+1G>A`	`chr11:59845374`	VCV439755	rs147785187	MIM:261000	CBLIF	Atlas
GJB2	`p.Met34Thr`	`chr13:20189481`	VCV17000	rs35887622	MIM:121011	GJB2	Atlas
VDR	`c.-1172A>G`	`chr12:47906043`	VCV3336650	rs4516035	—	VDR	Atlas
F5	`p.Arg534Gln`	`chr1:169549811`	VCV642	rs6025	—	F5	Atlas
PEX6	`p.Val788Met`	`chr6:42966044`	VCV556244	rs267608240	MIM:614862	PEX6	Atlas
GJB3	`p.Arg75Cys`	`chr1:34784985`	VCV297191	rs370476720	MIM:133200	GJB3	Atlas
SCN4A	`p.His599Arg`	`chr17:63961242`	VCV324537	rs187401185	MIM:168300	SCN4A	Atlas
C19orf12	`p.Gly42Arg`	`chr19:29708290`	VCV183298	rs200133991	MIM:614297	C19orf12	Atlas
LRP5	`p.Val233Glu`	`chr11:68363758`	VCV866038	rs1177484229	—	LRP5	Atlas
BLM	`p.Val724Ala`	`chr15:90765392`	VCV567134	rs750687788	MIM:210900	BLM	Atlas
ANK2	`p.Arg3906Trp`	`chr4:113373306`	VCV18059	rs121912706	MIM:600919	ANK2	Atlas
NOD2	`p.Arg675Trp`	`chr16:50712015`	VCV4693	rs2066844	MIM:266600	NOD2	Atlas